ECE 2305 – Introduction to C Programming
Programming Tutorial 27
Data Structures and Functions

Use the **`struct`** described in Programming Tutorial 26 and write a user defined function that uses the **`stuct`** Data Members. In the main function, prompt the user to enter the number of pieces of fruit to purchase. Write the function to take the fruit object (by reference) and the number of pieces to be purchased (by value) as parameters. The function shall update the quantity of fruit and return the purchase price.

Input

```
// ECE 2305 - Tutorial 27

#include <iostream>
#include <iomanip>
#include <string>

using namespace std;
struct Fruit {
    string name;
    int quantity;
    double price;
};
double purchaseFruit(Fruit& f, int numBought) {
    if (numBought > f.quantity) {
        cout << "Not enough " << f.name << " in stock.\n";
        return 0.0;
    }
    f.quantity -= numBought;
    return numBought * f.price;
}

int main() {
    Fruit apple = { "Apple", 10, 0.99 };

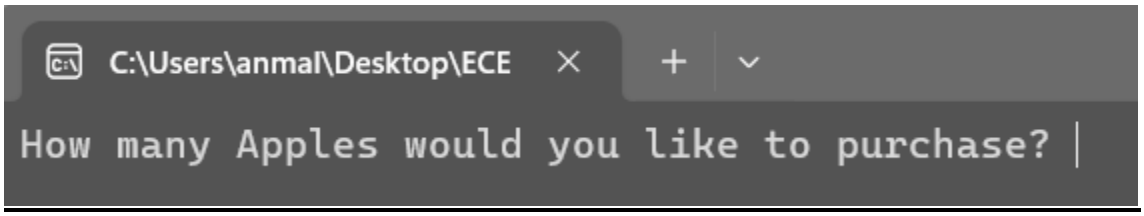
    int toBuy;
    cout << "How many " << apple.name << "s would you like to purchase? ";
    cin >> toBuy;

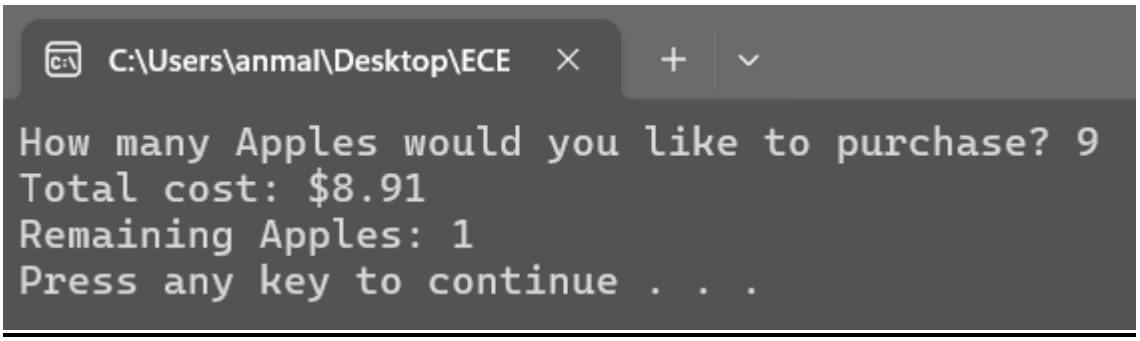
    double total = purchaseFruit(apple, toBuy);

    if (total > 0.0) {
        cout << fixed << setprecision(2);
        cout << "Total cost: $" << total << endl;
        cout << "Remaining " << apple.name << "s: " << apple.quantity <<
endl;
    }

    system("pause");
    return 0;
}
```

Output


C:\Users\anmal\Desktop\ECE
How many Apples would you like to purchase?


C:\Users\anmal\Desktop\ECE
How many Apples would you like to purchase? 9
Total cost: $8.91
Remaining Apples: 1
Press any key to continue . . .